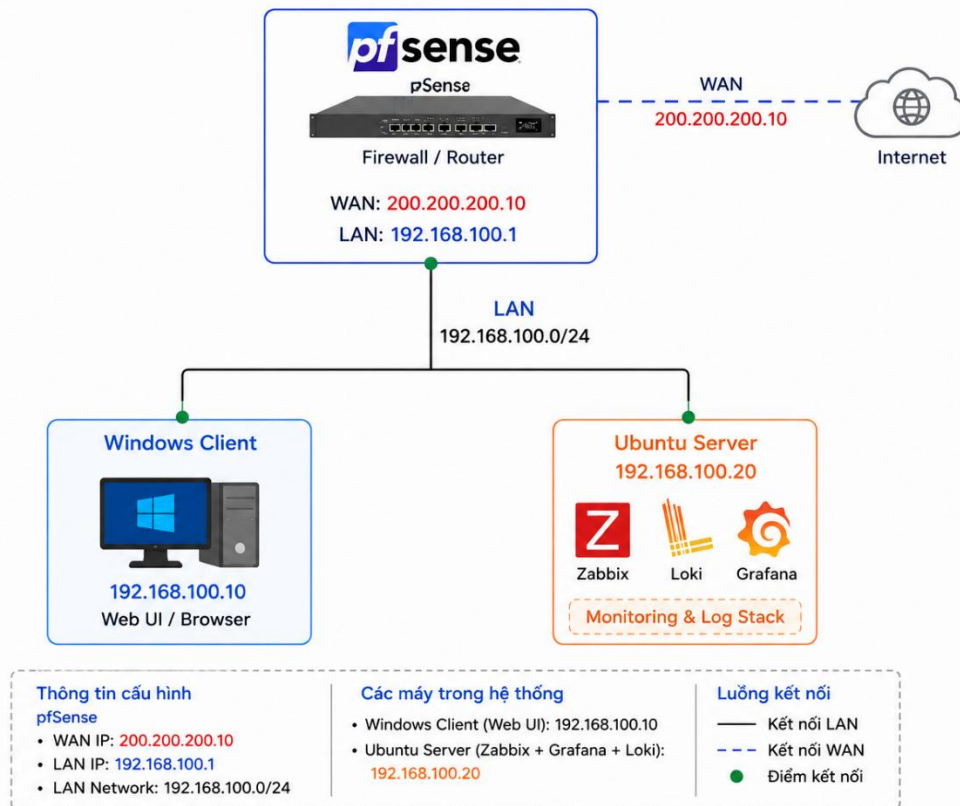


Cấu hình pfsense lấy log trên Grafana dùng loki



Cài Loki/Alloy

```
grep -R "grafana" /etc/apt/sources.list /etc/apt/sources.list.d/ 2>/dev/null
```

```
apt policy Grafana
```

```
apt policy loki
```

```
sudo apt install -y loki
```

```
zabbix@zabbix:~$ loki --version
loki, version 3.7.6 (branch: release-3.7.x, revision: 5003600d)
  build user:      root@31d37f362fe2
  build date:      2026-08-05T15:36:54Z
  go version:      go1.26.5
  platform:        linux/amd64
  tags:            netgo
```

cài Grafana Alloy

sudo apt install -y alloy

```
zabbix@zabbix:~$ alloy --version
alloy, version v1.18.1 (branch: HEAD, revision: 6012ec4)
  build user:      root@186280d16a5b
  build date:      2026-08-06T13:13:37Z
  go version:      go1.26.5
  platform:        linux/amd64
  tags:            gore2regex,netgo,embedalloyui,promptail_journal_enabled
```

cấu hình Alloy

```
zabbix@zabbix:~$ sudo cat /etc/alloy/config.alloy
[sudo] password for zabbix:
loki.source.syslog "pfsense" {
  listener {
    address      = "0.0.0.0:5514"
    protocol     = "udp"
    syslog_format = "rfc3164"

    labels = {
      job = "pfsense",
      host = "pfsense",
    }
  }

  forward_to = [loki.write.local.receiver]
}

loki.write "local" {
  endpoint {
    url = "http://127.0.0.1:3100/loki/api/v1/push"
  }
}
```

loki.source.syslog → UDP 5514 → nhận log pfSense → loki.write.local →
http://127.0.0.1:3100 → Loki

Start Alloy

```
zabbix@zabbix:~$ systemctl status alloy --no-pager
● alloy.service - Vendor-agnostic OpenTelemetry Collector distribution with programmable pipelines
   Loaded: loaded (/usr/lib/systemd/system/alloy.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-11 01:03:31 UTC; 6h ago
     Docs: https://grafana.com/docs/alloy
    Main PID: 995 (alloy)
      Tasks: 8 (limit: 2205)
    Memory: 77.6M (peak: 282.6M swap: 40.1M swap peak: 41.7M)
       CPU: 50.479s
    CGroup: /system.slice/alloy.service
            └─995 /usr/bin/alloy run --storage.path=/var/lib/alloy/data /etc/alloy/config.alloy
```

Trên pfsense

Status → System Logs → Settings

Remote Logging Options	
Enable Remote Logging	☒ Send log messages to remote syslog server
Source Address	Default (any) ▼ This option will allow the logging daemon to bind to a single IP address, rather than all IP addresses. If a single IP is picked, remote syslog servers must all be of that IP type. To mix IPv4 and IPv6 remote syslog servers, bind to all interfaces. NOTE: If an IP address cannot be located on the chosen interface, the daemon will bind to all addresses.
IP Protocol	IPv4 ▼ This option is only used when a non-default address is chosen as the source above. This option only expresses a preference; If an IP address of the selected type is not found on the chosen interface, the other type will be tried.
Remote log servers	192.168.100.20:5514 IP[:port] IP[:port]
Remote Syslog Contents	☒ Everything ☐ System Events ☐ Firewall Events ☐ DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns) ☐ DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client) ☐ PPP Events (PPPoE WAN Client, L2TP WAN Client, PPTP WAN Client) ☐ General Authentication Events ☐ Captive Portal Events ☐ VPN Events (IPsec, OpenVPN, L2TP, PPPoE Server)

pfSense → Zabbix đã ok

Alloy → Loki.

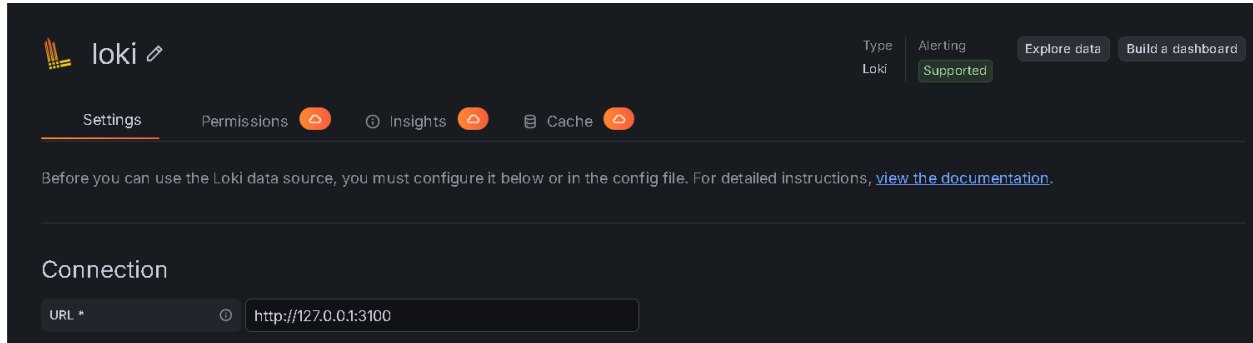
Kiểm tra Loki đã nhận label pfsense chưa

```
zabbix@zabbix:~$ curl -s http://127.0.0.1:3100/loki/api/v1/labels
{"status": "success"}
```

Connect tới Grafana

Connections → Data sources → Add data source → Loki

URL: http://127.0.0.1:3100

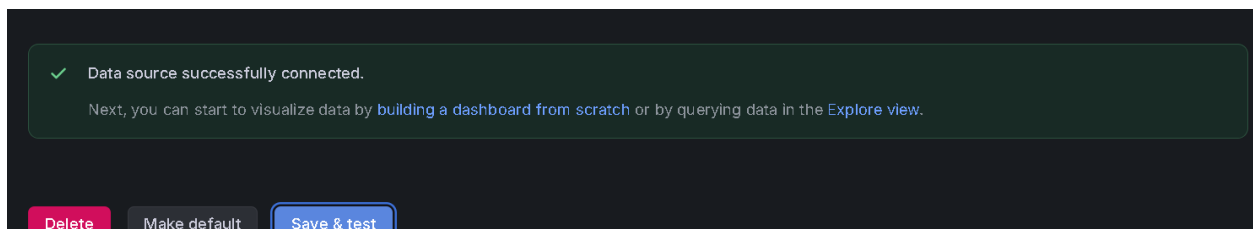


Check log trên Zabbix lấy từ pfsense

```
zabbix@zabbix:~$ curl -G -s "http://127.0.0.1:3100/loki/api/v1/query_range" \
--data-urlencode 'query={job="pfsense"}' \
--data-urlencode 'limit=20' | jq
{
  "status": "success",
  "data": {
    "resultType": "streams",
    "result": [],
    "stats": {
      "summary": {
        "bytesProcessedPerSecond": 0,
        "linesProcessedPerSecond": 0,
        "totalBytesProcessed": 0,

```

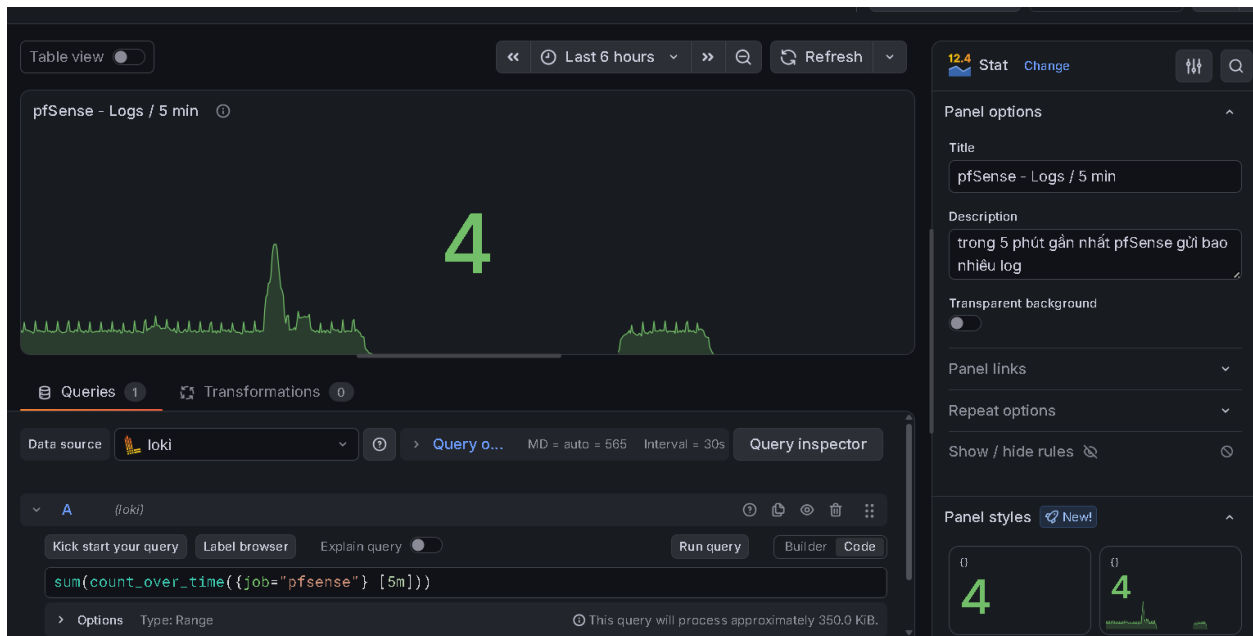
⇒ đã thành công hoàn toàn ở tầng pfSense → Alloy → Loki



⇒ Connect source thành công

Tạo dashboard check log

Check log mỗi 5 phút



Check lỗi mỗi 5 phút

